

Executive summary

Uptime Intelligence report: January 2026

Five data center predictions for 2026

In this report, Uptime Intelligence looks beyond the more obvious trends of 2026 and examines some of the latest developments and challenges shaping the industry. AI is transforming data center strategies, but adoption patterns remain uncertain due to technical and market immaturity, complex infrastructure requirements, and IT and infrastructure development timelines.

In 2026 and the years that follow, AI workload growth will remain concentrated among organizations that can support high-density deployments, while power demands will strain aging grids, raise failure risks and increase carbon emissions. These pressures will force some operators to reassess resiliency strategies and reconsider whether — and how — to reach net-zero commitments. At the same time, AI automation in data centers will transition from pilot projects into operations, delivering measurable efficiency gains.

Uptime Intelligence: actionable insight for the digital infrastructure ecosystem.

To access the entire *Five Data Center Predictions for 2026* report and Uptime Intelligence on an evaluation basis, please visit intelligence.uptimeinstitute.com/request-evaluation

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Uptime Institute serves all stakeholders that are responsible for IT service availability through industry-leading standards, education, membership, consulting and award programs delivered to enterprise organizations and third-party operators, manufacturers and providers.

Synopsis

The critical digital infrastructure sector continues to expand strongly, but it faces growing challenges related to power, infrastructure complexity and cost, and sustainability. At the same time, uncertainty about how AI will reshape demand is complicating capacity planning and resiliency strategies. This report explores these issues and their implications for the industry in 2026 and beyond, including the concentration of high-density AI infrastructure, power constraints that limit growth and strain grids, the growing relevance of carbon capture and storage, and the transition of AI automation from experimentation to operational deployment in the data center.

Summary of the data center predictions for 2026

1. Giants, supergiants and the rest: the AI ecosystem takes shape

AI compute and high-density infrastructure will increasingly concentrate among a small number of organizations. Enterprises and IT service providers will limit investment to inference and only some training, while relying on cloud platforms and pre-trained models. All data center operators will see some AI-driven demand, alongside trickle-down benefits in power and cooling technologies.

2. Developers will not outrun the power shortage

AI-driven load growth will intensify pressure on already constrained grids, widening the gap between data center and utility timelines. Short-term solutions, such as on-site generation and repurposed crypto sites, will become more expensive and less available. As power constraints intensify, operators will face higher costs, greater public opposition, and possible grid-interactivity requirements.

3. As emissions soar, operators look to carbon capture

The projected 75-125 GW growth in global data center power demand through 2030 will drive greater reliance on natural gas, while further complicating existing net-zero commitments. As a result, some operators will collaborate with energy developers to integrate carbon capture and storage (CCS) systems into new natural gas generation projects. Despite cost, siting and technology limits, many operators will consider CCS for the direct emissions reductions and lower offset costs.

4. Scale adds new challenges, but resiliency will still be essential

Rising grid instability and the scale of AI capacity will increase the risk of severe outages, forcing operators to re-examine the role of availability zones and to consider mixed levels of resiliency for different workloads. Investments in on-site generation and equipment redundancy will lead operators to share assets — requiring a more integrated view of resiliency.

5. AI automation in the data center moves from pilots to production

AI-driven automation within the data center will transition from experimental use to supporting daily operations. Reinforcement learning, hybrid digital twins, and early industrial copilots will support closed-loop optimization and operator decision-making, while rules-based systems handle routine workflows. Semi-autonomous operations will expand, but human oversight will remain essential.



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Introduction

At the beginning of each calendar year, Uptime Intelligence compiles a short list of trends or predictions that will be relevant to the digital infrastructure sector for the year (and years) ahead. These lists aim to highlight crucial, yet often overlooked industry topics, which encourage a closer examination. Our five predictions for 2025 have proved to be largely accurate and will remain relevant for 2026. These are summarized in **Table 1**.

This year's predictions build on those for 2025 and focus on the industry's continued growth and associated challenges, while also recognizing AI as a powerful, transformative accelerant to growth (see *Five data center predictions for 2025*). AI is also the key driver for a wave of investment that will underpin digital infrastructure for many decades — even if the speed and ultimate size of the build-out (and financial rewards) remain unclear.

AI has only recently become the dominant narrative in industry media headlines, and most operators are still planning in the face of uncertainty rather than executing on clearly AI-driven demand

Several of the factors shaping these predictions sit outside the direct control of data center operators and will take years to fully materialize. AI has only recently become the dominant narrative in industry media headlines, and most operators are continuing to plan in the face of uncertainty rather than executing on clearly AI-driven demand. This period of exploration and adjustment is likely to continue as organizations determine how AI fits into their business strategies over the longer term.

Importantly, this uncertainty around AI has not slowed the broader growth of critical digital infrastructure. Demand for general-purpose compute, storage and connectivity remains strong and is likely to persist regardless of the trajectory of AI applications. Even where AI investments fail to deliver near-term returns, the promise and demands of AI are having a catalytic effect, driving innovation in areas such as power delivery, cooling design and grid integration, which may strengthen the industry's long-term foundations.

Many long-standing barriers to growth remain unresolved. Power availability and reliability continue to constrain expansion in key markets; public opposition to new developments is increasing; and supply chain disruptions are adding complexity to development timelines. These challenges are compounded by rising costs and growing concerns over systemic failures as aging grids are strained beyond their limits. AI has amplified these pressures by compressing timelines and raising expectations.

Environmental considerations are also becoming more difficult to separate from growth strategies. As developers and operators pursue new approaches to secure reliable power, they face heightened regulatory scrutiny, public concern and trade-offs related to emissions. For many operators, this has renewed an interest in exploring carbon management approaches beyond traditional offsets as the industry seeks ways to balance reliability, expansion and sustainability.

Entering 2026, a clearer segmentation is emerging. The most demanding AI workloads are driving a distinct class of high-density infrastructure architectures with greater power and cooling requirements. The majority of data center capacity, however, continues to support more traditional digital services. This separation may ultimately benefit the industry by clarifying risks, investment priorities and operational strategies.

The strategies operators adopt to manage these constraints — across power, infrastructure and operations — will play a defining role in shaping the industry's trajectory in 2026 and the years ahead.

Table 1

Predictions for 2025 recap

2025 prediction	Summary
Data center resource use will raise deep questions — and opposition	Governments will support rapid expansion of the data center industry, but climate commitments will be downgraded or postponed as optimism around AI continues to drive growth.
Most AI models will be trained in the cloud	Most investments in AI infrastructure for large-scale training will come from hyperscalers and cloud providers while enterprises rely on public cloud services and pre-trained foundation models, fine-tuning them to reduce computational overhead.
Grid demand will require active participation from data centers	New and expanded data centers will increasingly be expected to provide, store, or shed power to support grids, with non-latency-sensitive workloads reducing power use when required.
AI to trigger radical overhaul of data center electrification	Next-generation AI requirements will push operators to explore new power architectures, leading to early adoption of medium-voltage distribution and solid-state transformers.
Nvidia's vision for data centers is not without alternatives	Nvidia's near-monopoly in enterprise GPUs, driven by performance but limited by cost and scarcity, will encourage some organizations to seek alternatives and contribute to greater AI hardware diversity.

PREDICTION 1

Giants, supergiants and the rest: the AI ecosystem takes shape

Key trends

- AI compute and the required high-density infrastructure will become increasingly concentrated among a small number of organizations and their close partners.
- Enterprises and IT service providers will increase investment in inference and some training workloads. However, they will avoid investing in large-scale high-density infrastructure, relying largely on cloud services and pre-trained foundational models.
- All data center operators will see some increases in demand from AI-related services. They will also benefit from emerging options in power distribution and honed liquid cooling systems as equipment manufacturers spend more on research and development.

The world of generative AI is one of extremes. On the one hand, hundreds of billions of dollars will continue to be spent on infrastructure for generative AI workloads. In the process, these infrastructures are becoming like heavy industry, consuming land, water, energy and supply chain resources at an unprecedented rate. On the other hand, enterprise investment in AI compute infrastructure has been decidedly measured. IT analyst and investments firm studies put spending on AI by enterprise IT at only 3-5% of their annual budgets from 2023 to 2025.

This disconnect is not a contradiction, and it is not about leaders and laggards. Most organizations — even those enthusiasts in the power of AI — can adopt pre-trained models and feed it their own data while retaining sufficient control over the model and IT security.

Those enterprises (and their IT service providers) that require AI models to be tailored to specific domain and business needs will deploy relatively small amounts of capacity, and mostly at moderately high rack densities (under 60 kW).

Demand for compute and electrical power for inference remains uncertain. The expectation that infrastructure for inference, due to broad adoption of AI-powered services, will outstrip training in size and power is likely misguided. The capacity needs of enterprise inference are incremental and large training-first systems will serve much of the inference workloads.

Over the next few years, Uptime Intelligence expects AI compute to become even more concentrated at the high-end of the market, not less. AI compute is becoming supercomputer-style, featuring rack densities of hundreds of kilowatts and systems reaching megawatts.

Much of AI supercomputing will take place in giant data center campuses, and also in numerous metro-area colocation sites that run localized training, copies of foundational models and AI infrastructure services for enterprises. The rest of AI infrastructure will remain large and diverse, with a growing number of organizations having high-density racks, but most organizations will invest relatively cautiously.

Crucially, AI infrastructure concentration will bring diffusion of new facility technologies. The new power and cooling equipment designed to address the challenges of operating large and ultra-dense AI facilities will prove valuable across the broader industry.

AI approaches infrastructure singularity

The battle to develop powerful and widely applicable AI models has become a high-stakes competition — akin to a space race — that only a few will be able to sustain over time.

The vast investments being made are driven not only by opportunity, but also by fear. This investment is fueled by the widespread belief that AI is a winner-takes-all market, which threatens the existence of lucrative adjacent markets from web search to social media to enterprise software. Factors that will shape AI infrastructure in the coming years include:

- **Incessant obsolescence.** The need to produce the best solution in generative AI, even marginally, comes at a high cost. The scaling laws of generative pre-trained transformer (GPT) architectures dictate that training significantly better AI models requires disproportionately more computational budget. AI developers take order-of-magnitude leaps between model generations to achieve significantly better model behavior. AI training systems today are a thousand times more performant than 3 years ago. This technological push makes 3-4-year-old hardware obsolete.

Few enterprises will be able to justify the investment needed to keep up with this kind of AI development. Even fewer will do it using their own hardware (due to data governance or cost reasons). Those that do invest will deploy moderate densities for their, often air cooled, IT.

- **AI compute is supercomputing.** AI data centers are getting larger — much larger — concentrating compute power to fewer sites and organizations. This is evident in the rise of “supergiant” data center campuses being built by hyperscalers and major AI model owners, as well as in multi-megawatt GPU clusters, mostly hosted by colocation providers in metro-area facilities.

The construction of giant campuses demonstrates how AI training is driving up the scale of the compute engines to supercomputer levels. Uptime Intelligence identified nearly 30 new campus proposals in 2025 that surpassed a gigawatt, along with almost 100 additional projects in the hundreds of megawatts range worldwide — on top of the 200 previously announced.

Even if most projects do not reach their planned capacity, they will still account for tens of gigawatts which are planned for AI training workloads, with inferencing a secondary workload.

- **Enterprise AI inference remains relatively small.** Even large enterprise inference workloads, supporting many thousands of concurrent customers or internal users, run well on a few servers using GPUs or AI-inference silicon. For example, a large retail bank serves millions of customers and thousands of employees with advanced LLM bots using fewer than 100 GPUs across multiple sites, which includes spare capacity for resiliency. The same bank has tens of thousands of servers overall.

Most businesses will have much lower requirements. Although enterprise inference workloads will accumulate globally, they will not represent a substantial uplift relative to other enterprise applications. But the more significant point is that inference demand from businesses accounts for only a small fraction of the millions of high-performance GPUs installed globally each year.

AI tidal forces stretch the industry

Equipment manufacturers will introduce new power and cooling systems to meet the needs of AI supercomputing infrastructure as it diverges from generic data centers. Unlike pre-existing scientific and technical supercomputing, AI supercomputing has grown large enough to economically sustain the parallel technical development of new facility equipment designed for both large-scale and ultra-dense data centers.

These expanded research and development efforts will benefit the wider industry, offering data center operators new technical options for both new builds and facility refurbishments. The foremost area where this will occur is electrification, but cooling is also set to benefit.

Electrification

The push for densification drives the development of new products, such as high-capacity busways, switches, breakers, transformers and power distribution units, but the major shift will come from new electrical topologies.

A key focus is in bringing medium-voltage (MV, typically 11 kV or higher) facility distribution closer to the IT load, and the re-introduction of direct current (DC) power. Data center electrification vendors are developing a slate of new products, which were widely discussed in 2025, to enable new electrification topologies in 2026 and beyond.

These developments include new MV UPS systems, 800V DC UPS systems and solid-state transformers that promise to make the adoption of the former two less expensive and complex. Higher distribution voltages help reduce conductor sizes, which in turn also opens up the possibility of longer run lengths to adapt the layout of a data center site. DC distribution specifically will allow for simpler power electronics in IT systems and improve overall energy performance. These advantages will apply to all large data centers (including cloud) — not just high density.

Cooling

Data center operators are gaining design and operational experience with both high-density air cooling and direct liquid cooling (DLC) systems, including how to manage the design and operational practices in flexible, hybrid environments.

This will lead to a greater selection of products, including coolant distribution units, fluid networks, cold plates and immersion vats. Even more importantly, this will lead to much broader support for DLC options from major IT OEMs — the lack of options has been a major constraint up to now.

The largest AI data centers, however, will be increasingly distinct. The duality of data center infrastructure, characterized by markedly different design power density targets, will benefit the entire data center industry through stronger economies of scale and a richer set of facility topology options, as well as choices in electrical and mechanical products. Operators of generic data centers will also have a wider range of options to consider, even though their design power densities will be moderate and their prime objectives will remain more focused on cost and resiliency.

PREDICTION 4

Scale adds new challenges, but resiliency will still be essential

Key trends

- Fears of severe and systemic failures are mounting, driven by growing grid instability. The scale and concentration of new data center capacity are compounding the risks.
- As AI workloads become more established, operators may be forced to re-examine the role of availability zones and consider mixed levels of resiliency for different workloads.
- Large investments in on-site generation and equipment redundancy will likely lead to more operators sharing their assets, requiring a more integrated view of resiliency.

For decades, the industry has agreed on the core principles of physical infrastructure resiliency: while it is not possible to avoid failures altogether, the risk and impact of an outage can be minimized through well-established design and operational practices. Notably, these include independence from the power grid and other utilities, and redundancy in critical systems to achieve concurrent maintainability of the facility (i.e., zero scheduled downtime). Further design and operational measures aim to increase facility tolerance against failures.

The industry shares this resiliency philosophy because it has proven effective. According to Uptime Institute survey data (see *Annual outage analysis 2025*), outage rates per site — and certainly relative to IT load — have been trending downwards over time.

Yet, over the next few years, Uptime Intelligence expects resiliency to re-emerge as a major area of focus and concern for the industry. The reasons for this include:

- Structural, long-term challenges with many power grids, coupled with the concentration and scale of data centers are increasing the likelihood of grid instability and failures. Increased regulatory intervention is also likely.
- The rise of AI is changing the load profile, behavior and distribution of IT, making strategies such as availability zones (AZs) and distributed resiliency more difficult or more expensive to achieve.
- Growing reliance on expensive, large-scale on-site power generation will force some operators to share their generating capacity. This will require closer collaboration with transmission service operators (TSOs) to manage risks and capacities more actively, and to balance operators' resiliency goals with the needs of their grid partners.

All this is happening against a backdrop of increasing jeopardy. While large outages have become less likely relative to the size of the industry, the consequences of a failure have risen — a trend reflected in Uptime Intelligence data (see *Uptime Institute Global Data Center Survey 2025*). The ripple effects of a major site outage can extend far: the loss of a core cloud or software service can affect thousands of third-party applications and services, while the automated rerouting of traffic may result in further cascading service failures due to network congestion.

Regulators have noticed these risks to national economies. Some have designated data centers as critical national infrastructure as an initial step toward more intervention. Others are planning new resiliency-focused legislation.

Concentration and scale

The regional concentration of IT service providers and fiber network operators was once a major concern for the resiliency of the internet, but this has now largely passed. However, high concentration has become a risk for data center regions themselves, such as Northern Virginia (US) or the Dublin (Ireland) area.

There are three specific concerns. First, a combination of growing electrification, aging transmission equipment, and an increasing share of intermittent generation in the energy mix are creating conditions for grid instability. In September 2025, the US Department of Energy (DoE) warned that grid power outages could increase by a factor of 100 by 2030.

Most data centers are designed to handle power outages by switching to engine generators. However, more frequent grid disturbances and outages will place greater stress on critical power systems. A major grid failure can also lead to refueling delays, increased particulate emissions in populated areas, and operational problems caused by loss of communications and staff shortages.

Most data centers are designed to handle power outages by switching to engine generators. However, more frequent grid disturbances and outages will place greater stress on critical power systems

Second, data centers have become part of the problem. Large campuses and concentrations mean that data centers are stretching grid supply, and their behavior can affect grid stability (see *Power companies act to stop data center-induced blackouts*). Two examples serve as a warning: voltage fluctuations in Northern Virginia in July 2024 and in Ireland in December 2022 led to many data centers disconnecting from the grid near simultaneously, causing power surges (oversupply) that almost resulted in blackouts in both cases.

Heeding these lessons, authorities and TSOs in several countries — including the US, Australia, Singapore, the UK and much of Europe — are now working on new grid connection rules. These will oblige operators to tolerate higher voltage fluctuations and to avoid, or slow down, load disconnection during grid disturbances.

The third concern is that major grid outages may strain, or even cripple, fuel supplies. During the Spanish blackout in early 2025, Airbus's Madrid data center experienced a critical diesel shortage as fuel suppliers became overwhelmed. A major power outage affecting London Heathrow in the UK also caused some operators to review their access to fuel. Storing larger on-site fuel reserves is usually the first option, but operators are increasingly focusing on their arrangements for emergency resupplies. In some jurisdictions, critical national infrastructure classification will help secure priority access to fuel during large-scale outages.

IT resiliency: cost calls for more dynamism

For decades, the practice of high-availability and fault-tolerant clustering across multiple data centers has been used for select mission-critical applications. The emergence of cloud AZs, together with affordable IT hardware and novel application development styles, has popularized the idea of multi-site resiliency for a much broader range of applications. Increasingly, new IT applications are operated in a scale-out fashion, distributed across several facilities using load balancers.

However, there are challenges: software design complexity, the difficulty of mapping and eliminating single points of failure, all coupled with the cost of wide-area networking required for high-speed data replication (see *Cloud availability comes at a price*). Distributed resiliency techniques are neither ubiquitous nor bulletproof, but they have proven successful for services designed for the cloud era. In 2025, IT and distributed resiliency increasingly complemented site-based resiliency and, as these techniques become more robust, they are progressing toward replacing it in some use cases.

Generative AI compute is likely to bifurcate this dynamic. Capacity redundancy for computationally intensive AI workloads, such as large-scale inference, is expensive to begin with, and AI-dependent workloads are therefore attracting renewed focus on hardware and physical infrastructure resiliency. For AI training, distributed resiliency is a non-starter.

Depending on their risk tolerance and IT architectures, some organizations will continue to absorb the costs of high physical infrastructure resiliency across all applications, including AI compute. Others, however, will opt to fully examine what level of resiliency each application or service really needs, starting with larger AI workloads. As analytical tools and data improve, making these distinctions may become more practical than in the past.

Large dense sites will pay for resiliency

Since 2021, Uptime Intelligence has identified plans for more than 300 data center campuses worldwide exceeding 100 MW — some even surpass 1 GW. Hundreds more are in development at a smaller scale, although these are still very large by historical standards.

Providing full-sized emergency backup power at this scale entails enormous cost, often amounting to tens of millions of dollars to install. However, capital costs are only part of the problem. The process of testing and maintaining engines and fuel systems is increasingly laborious. Then there is the logistical issue of resupplying fuel at such a large scale.

In the battle between OpenAI, Google, Meta and others to train the best-performing generative AI models, resiliency has not been a high priority — moving fast and breaking things is the Silicon Valley ethos. This has promoted a perception that AI training facilities running batch loads can be built without concurrent maintainability or even emergency backup power.

However, this behavior is unlikely to be followed by many new data center builders, despite the expense and challenges of higher resiliency levels. Uptime Institute is aware of, and in some cases involved in, the construction of large facilities that do not compromise on resiliency.

Some larger facilities will likely build their own (or colocate with) prime power generation and distribution infrastructure in response to limited grid power. Adding redundancy to prime generation will cost less, relatively, than providing standalone backup power, particularly since some of this capacity can be sold to the grid as part of demand response schemes. In any case, IT infrastructure costs will dwarf the cost of added power resiliency.

This is underlined by a second, broad-based observation. Except for a small number of very high-density, HPC-like facilities, most data centers will likely support, or need the capability of supporting, a rich mix of workloads. Because technical requirements several years into the future are highly uncertain, operators will want to ensure that facility investments remain durable. Most data center customers, including hyperscale and other IT service providers, will continue to require strong resiliency guarantees.

Find out more

We hope you found this executive summary of our *Five Data Center Predictions for 2026* report valuable.

To access the entire *Five Data Center Predictions for 2026* report and Uptime Intelligence on an evaluation basis, please visit intelligence.uptimeinstitute.com/request-evaluation

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Uptime Institute is the Global Digital Infrastructure Authority. With over 4,000 awards issued in over 122 countries around the globe, and over 1,100 currently active projects in 80+ countries, Uptime has helped tens of thousands of companies optimize critical IT assets while managing costs, resources, and efficiency. For over 30 years, the company has established industry-leading benchmarks for data center performance, resilience, sustainability, and efficiency, which provide customers assurance that their digital infrastructure can perform across a wide array of operating conditions at a level consistent with their individual business needs. Uptime's Tier Standard is the IT industry's most trusted and adopted global standard for the design, construction, and operation of data centers. Offerings include the organization's Tier Standard and Certifications, Management & Operations reviews and assessments including SCIRA-FSI financial sector risk assessment, the Sustainability Assessment, and a broad range of additional risk management, performance, availability, and related offerings. Uptime Education training programs have been successfully completed by over 100,000 data center professionals, such as the much-valued ATD (Accredited Tier Designer) and AOS (Accredited Operations Specialist). The Uptime Education curriculum has been expanded by the acquisition of CNet Training Ltd. In 2023.

Uptime Institute is headquartered in New York, NY, with offices in London, Sao Paulo, Dubai, Riyadh, and Singapore, and full-time Uptime professionals based in over thirty-four countries around the world.

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